

Physical Science Part 4

1. The potential energy of a body at a certain height is 100 J. The kinetic energy possessed by it when it just touches the surface of the earth is _____.
 - A. greater than the potential energy
 - B. less than the potential energy
 - C. equal to the potential energy
 - D. cannot be known
2. What happens to the kinetic energy of an object if its speed is doubled?
 - A. It will be halved.
 - B. It will be doubled.
 - C. It will be tripled.
 - D. It will be quadrupled.
3. What is the type of energy possessed by a simple pendulum when it is at the mean position?
 - A. Kinetic energy
 - B. Mechanical energy
 - C. Potential energy
 - D. Sound energy
4. A ball is thrown upward into the air with a speed that is greater than terminal speed. On the way up, it slows down and after its speed equal the terminal speed but before it gets to the trajectory _____.
 - A. it speed up
 - B. its speed is constant
 - C. it continues to the slow down
 - D. its motion becomes jerky
5. When a bullet is fired from a rifle, the force in the rifle is equal to the force on the bullet. Why is it that the energy of the bullet is greater than the energy of the recoiling rifle?
 - A. Because there is more impulse on the bullet
 - B. Because the force on the bullet acts for a longer time
 - C. Because the force on the bullet acts for the longer distance
 - D. Because the bullet's momentum is greater than that of the rifle
6. Car A has a mass of 1000 kg and the speed of 60 kph while car B has a mass of 200 kg running at a speed of 30 kph. What is the kinetic energy of car A?
 - A. Half that of car B
 - B. Equal that of car B
 - C. Twice that of car B
 - D. Four times that of car B
7. An object is thrown straight up. Which of the following is TRUE about the sign of work done by the gravitational force while the object moves up and down?

- A. Work is zero on the way up, work is zero on the way down.
 - B. Work is negative on the way up, work is negative on the way down.
 - C. Work is negative on the way up, work is positive on the way down.
 - D. Work is positive in the way up, work is negative on the way down.
8. What happens to the kinetic energy of a moving object if the net work done is positive?
- A. The kinetic energy remains the same.
 - B. The kinetic energy increases.
 - C. The kinetic energy decreases.
 - D. The kinetic energy is zero.
9. Object A with a mass of 4 kg is lifted vertically 3 m from the ground, object B with a mass of 2 kg is lifted 6 m up. Which of the following statement/s is/are TRUE?
- I. Object A has greater potential energy since it is heavier.
 - II. Object B has greater potential energy since it is lifted to a higher position.
 - III. Two objects have the same potential energy.
- A. I only
 - B. II only
 - C. III only
 - D. not enough information given
10. According to the scientific definition of work, pushing a table accomplishes no work unless there is _____.
- A. an opposing force
 - B. a net force greater than zero
 - C. an applied force greater than its weight
 - D. movement in the same direction as the force
11. A machine does 2 500 J of work in one minute. What is the power developed by the machine?
- A. 21 W
 - B. 42W
 - C. 2 500 W
 - D. 150 000 W
12. To which type of simple machine does a pencil sharpener belong?
- A. Inclined plane
 - B. Lever
 - C. Pulley
 - D. Wheel and axel
13. Which of Newton's law best explains why motorists should always wear seat belts?
- A. Law of Inertia
 - B. Law of Acceleration
 - C. Law of Interaction
 - D. Law of Gravitation

14. The same net force is applied to object 1 and object 2. The observed acceleration of the two objects are not the same; object 1 has acceleration three times that of object 2. Which of the following is correct?

- A. Object 1 has more friction than object 2.
- B. Object 1 has one-third the mass of object 2.
- C. Object 1 has three times the mass of object 2.
- D. Object 1 has a different, less streamlined shaped than object 2.

15. The law of inertia applies to _____.

- A. moving objects
- B. objects that are not moving
- C. both moving and non-moving objects
- D. none of these

16. Which statement is CORRECT about the Law of Conversation of Energy?

- A. The total amount of energy is constant.
- B. Energy cannot be used faster than it is created.
- C. The supply of energy is limited so we must conserve.
- D. Energy must be conserved and you are breaking a law if you waste energy.

17. Which of the following can transform electrical energy to mechanical energy?

- A. Electric bulb
- B. Electric fan
- C. Loud speaker
- D. Television

18. Celso is making a picture frame. He sticks a nail into a piece of wood and hits the nail with a hammer. Which form of energy is NOT released when the hammer hits the nail?

- A. Chemical
- B. Kinetic
- C. Sound
- D. Thermal

19. Butch throws a baseball into the air. It rises, stops when it reaches its greatest height, and the falls back to the ground. At what point is kinetic energy initially converted to potential energy?

- A. When the baseball is rising
- B. When the baseball is falling
- C. Just after the baseball hits the ground
- D. While the baseball is stopped in the air

20. Which energy conversation takes place when a toaster is switched on?

- A. Chemical to sound
- B. Chemical to thermal
- C. Electrical to sound

D. Electrical to thermal

21. When coal is burned to produce electricity, the electrical energy produced is less than the potential energy of the coal. Which BEST explains this observation?

- A. The amount energy in the coal is not known.
- B. As the coil is heated, the molecules move so fast that they are destroyed.
- C. Some of the potential energy in the coal is converted into forms of energy other than electricity.
- D. Some of the potential energy in the coal requires more intense heat so that can be compared to electricity.

22. What is the TRUE of all combinations of resistors arranged in parallel?

- I. Current splits down each leg of the combination.
 - II. Voltage is constant across each leg of the combination.
 - III. Resistance is the same for each leg of the combination.
- A. I and II only
 - B. I and II only
 - C. II and II only
 - D. I, II and III

23. What is the closed, continuous path through which electrons can flow?

- A. Charge
- B. Circuit
- C. Resistor
- D. Voltage

24. What will be your charge if you scrape electrons from your feet while scuffing across the rug?

- A. Negative
- B. Neutral
- C. Positive
- D. Not enough information given

25. How will you force charge if the distance between the two charge spheres is doubled?

- A. The force is doubled.
- B. The force is tripled.
- C. The force is quadrupled.
- D. The force is multiplied eight times.

26. A 6 ohm and 12 ohm resistors are connected in parallel with each other to a battery. Which of the following is TRUE?

- A. More current will flow through the 20 ohm resistor.
- B. The current flowing through the resistor is the same.
- C. The resistor values must be equal for the circuit to work.
- D. The potential difference across the two resistors is the same .

27. A new 1.5 V dry cell furnishes 30 A when short-circuited. Find the internal resistance of the cell.

- A. $0.05\ \Omega$
- B. $0.10\ \Omega$
- C. $1.5\ \Omega$
- D. $30\ \Omega$

28. If three identical lamps are connected in parallel to a 6 V dry cell, what is the voltage drop across each lamp?

- A. 2 V
- B. 3 V
- C. 6 V
- D. 18 V

29. Which of the following describes an electric current?

- A. Regulates the movement of electric charge
- B. The opposition material offers to the electron flow
- C. The electric pressure that causes charged particles to flow
- D. The movement of charged particles in the specific direction

30. A 10 W bulb and a 20 W bulb are connected in a series circuit. Which bulb has the more current flowing in it?

- A. The 10 W bulb
- B. The 20 W bulb
- C. Both bulbs have the same current
- D. Cannot be determined

31. Which of the following are characteristics of a series circuit?

- I. The current that flows through each resistance is the same as the total current throughout the circuit.
 - II. The sum of all the individual voltage drops in the same as the applied voltage.
 - III. The reciprocal of the total resistance is equal to the sum of the reciprocal of the separate resistances.
 - IV. The voltage drop in each resistance is the same as the magnitude of the applied voltage.
- A. I and II only
 - B. I and III only
 - C. III and IV only
 - D. I, II, and III only

32. Which states that the sum of all potential rises and potential drops in any loop (closed circuit) is zero?

- A. Coulomb's Law
- B. Kirchoff's Law
- C. Law of Electrolysis
- D. Ohm's Law

33. A television and an electric fan are connected in parallel to a 220 V line. The current in the television is 8 A and in the electric fan is 3 A. What is the current flowing in the line?

- A. 2.67 A
- B. 5 A
- C. 11 A
- D. 24 A

34. What happens to the current if resistance increases?

- A. Decreases
- B. Increases
- C. No change
- D. Cannot be determined

35. Which of the following statement DOES NOT represent Ohm's Law?

- A. Current/potential difference = constant
- B. Potential difference/current = constant
- C. Potential difference = current x resistance
- D. Current = resistance x potential difference

36. An iron nail can be turned into a temporary magnet if it is held in a strong magnet field. Which method of magnetization is this?

- A. Charging
- B. Conduction
- C. Introduction
- D. Saturation

37. What is the resistance of an electric bulb drawing 1.2 A current and 6.0 V?

- A. $0.2 \, \Omega$
- B. $0.5 \, \Omega$
- C. $2 \, \Omega$
- D. $5 \, \Omega$

38. What happens to the current-carrying ability of a wire as temperature increases?

- A. There is no change.
- B. The wire can't carry any current.
- C. The wire can carry more current.
- D. The wire can carry less current.

39. A 300 W at 220 V TV set is used at an average of 4 hours a day for 30 days. If the electrical energy costs Php 1.60 per kilowatt-hour, how much is the energy cost?

- A. Php 5.76
- B. Php 22.50
- C. Php 57.60
- D. Php 225.00

40. Electromagnetic radiation can be used for wireless communications. Which of the following can be used to transmit signals such as mobile phone calls?

- A. Infrared
- B. Gamma
- C. Microwaves
- D. Radiowaves

Answer Keys:

1. B
2. D
3. A
4. C
5. C
6. C
7. C
8. B
9. C
10. D
11. B
12. D
13. A
14. B
15. C
16. A
17. B
18. A
19. A
20. D
21. C
22. A
23. B
24. C
25. C
26. D
27. A
28. C
29. D
30. C
31. A
32. B
33. C
34. A
35. D
36. C
37. D
38. D
39. C
40. C

